

Antibubble formation with Plateau borders

Ann Author,¹ Second Author,^{1,*} Charlie Author,^{2,3} and Delta Author^{1,†}

¹*Authors' institution and/or address*

*This line break forced with *

²*Second institution and/or address*

*This line break forced with *

³*Third institution, the second for Charlie Author*

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Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

Introduction

Antibubble is the twin brother of soap bubble. While bubble encases gas within a liquid shell, antibubble forms in soapy liquids and traps liquid within a spherical air shell. Naturally, its unusual configuration promises its unique applications, like ultrasound imaging, drug delivery, double emulsion and other possible areas [1–3]. But unlike the extensive research on bubbles, antibubbles are far less studied, as there aren't so many efficient ways to generate antibubbles.

Traditional generation methods focus on horizontal liquid interfaces, like injecting a liquid column[4] or droplet pairs [5] directly onto a horizontal pool of soapy water, or through a thin soap film [6]. Their common prerequisite is that the liquid had a We number ($\sim 10^2$) big enough to entrain air into the water. Yet their productivity is about 20% from our lab due to the easy drainage of air film.

A recent method generates antibubble through a natural convex interface [7], i.e., the upper node of the Plateau border (PB) [see Fig. 1(a)]. This structure will naturally seal more air, therefore putting off the drainage process. Here we've reached an unexpected productivity of about 80%, with We mostly (~ 10)! Besides, we report that changing ambient air pressure might lead to unexpected drainage during antibubble formation. The dynamics of the generating process are revealed with theoretical analysis and scaling laws are proposed to determine different criteria for droplet entering and sliding through PB.

Experimental Setup

The experimental setup is schematically shown in Fig.1 and is briefly described below. A 3D-printed triangular prism frame (photosensitive resin, $a = 30$ mm and $b = 45$ mm) is first dipped into a soap solution of density $\rho = 1.0 \times 10^3$ kg/m³ and surface tension $\sigma = 30.2$ mN/m (pendant drop method), composed of 5.6% wt of commercial detergent, 5.6% wt of glycerol and 88.8% wt of ultrapure water. When the frame is subsequently extracted and fixed vertically above the liquid pool, a well-defined PB ($w \approx 90$ μ m, $l \approx 33.6$ mm, and $\alpha \approx 109.5^\circ$) forms instantly. We then inject droplets (radius $R = 1.0 - 2.2$ mm equivalent with that of the same droplet volume as a sphere) of the same solution through a syringe pump from a distance h right above the upper node of the PB. By varying h , a range of the impact velocity $u = 0.1 - 1.8$ m/s of the droplet is **acquired**.

Besides, we place the whole setup into a vacuum box with ambient pressure $p = 0.1 - 1.0$ atm. The interaction between the droplet and the border is recorded by a high-speed camera (v711, Phantom, U.S.A.) with 3000 to 10000 frames per second from the side view.

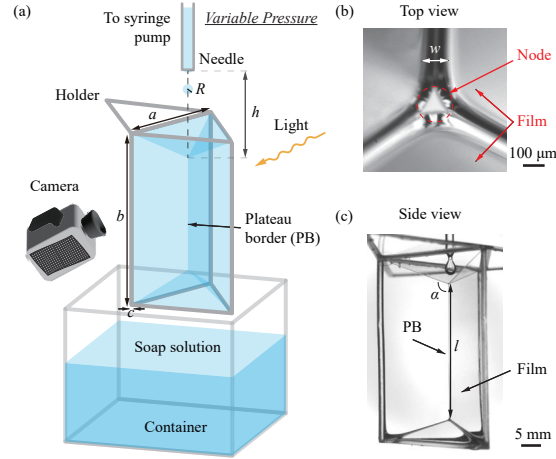


FIG. 1. Experimental configuration and notation (The vacuum box is omitted). (a) Schematic of the experimental setup. (b) Top view of the upper-node of the Plateau border. (c) Photo of the test apparatus from side view.

Results and discussion

Fig.2 shows three typical behaviours of droplets with different impact velocity u and radius R . Here $t = 0$ indicates the time at which each droplet begins to deform the soap film. When u and R are relatively low, the droplet may coalesce with the soap film above or even bounce back, leading to results named "outer coalescence" (Fig.2(b), $R = 1.13$ mm, $u = 0.40$ m/s, 33). When u or R increases to some extent, the droplet will be able to pass through (penetrate) the upper node of the PB. Basing on the maintenance or collapse of the air film in the multi-layer structure after pinch-off, the last two categories are named "packing" (Fig.2(a), $R = 1.75$ mm, $u = 0.18$ m/s, 44) and "inner coalescence" (Fig.2(c), $R = 1.16$ mm, $u = 0.43$ m/s, 24).

In the case of "packing", the droplet gains enough velocity after free fall and therefore successfully deform the film into a pocket without bouncing or coalescence. The concavity of the film structure helps the droplet to entrap air from both above and below. This pocket then pinches off ($t = 26.7$ ms), forming a stable multi-layer structure (1G1L, i.e. one gas film plus one liquid film) at the top of the PB. The droplet then transports in the PB under gravity and shows a shape oscillation due to capillary waves caused by the impact. When the droplet falls to the bottom of

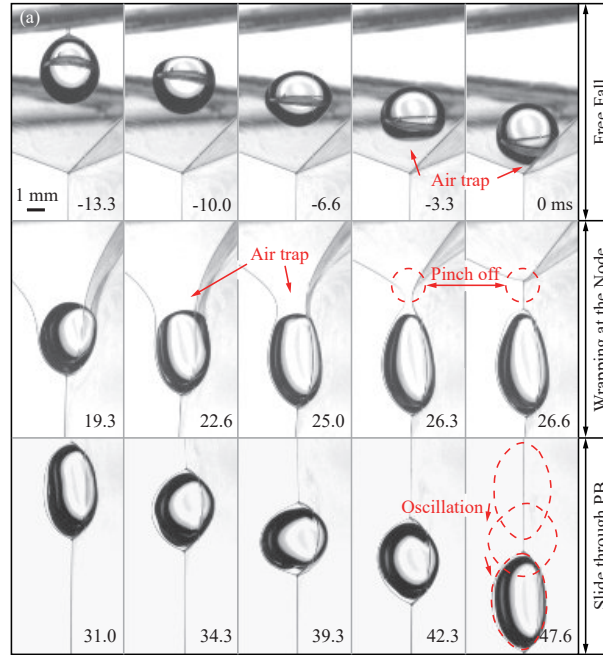


FIG. 2. Image sequences showing three typical behaviours of droplets with different impact velocity u and radius R for (a) "Packing", $R = 1.75$ mm, $u = 0.18$ m/s and $We = 1.94$; (b) "Outer coalescence", $R = 1.13$ mm, $u = 0.40$ m/s; (c) "Inner coalescence", $R = 1.16$ mm, $u = 0.43$ m/s; (d) Antibubble formation after the droplet passes through the Plateau border. $R = ?$ mm, $u = ?$ m/s.

the PB, the lower node opens and a packed droplet with a thin air layer (see the enlarged snapshot) is separated. An antibubble is then formed when the packed droplet impacts onto the soap solution below; see Fig.2(d).[]].

With relatively less u or R , droplets of other two categories will exhibit similar behaviours except for their *destinies*. The air gap in between will collapse either before or after pinch-off, finally leading to the coalescence between droplets and the film; see Fig.2(b)-(c). These cases will leave no packed droplets to impact onto the soap solution, and hence no antibubbles will emerge. So it's important to find a parameter within which we can generate antibubbles through a well-shaped PB.

Conclusions

We also found that the PB micro channel is capable of transporting other kinds of liquids, which might give rise to the development of drug delivery industry.

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* Second.Author@institution.edu

† CorrespondingAuthor@mail.tsinghua.edu.cn

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